

Seasonal patterns and preterm birth: a systematic review of the literature and an analysis in a London-based cohort

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Accepted 18 July 2006. Published OnlineEarly 15 September 2006.

Objective The objectives of this study included a systematic review of the countries in which a seasonal pattern of preterm birth has been reported and an analysis on the seasonal variability of preterm birth in a London-based cohort.

Design Cross-sectional study.

Setting Eighteen maternity units in a London health region from 1988 to 2000.

Population The study population comprised 482 765, live singleton births born after 24 weeks of gestation and weighing more than 200 g.

Methods Systematic review and secondary analysis of seasonality over 13 years of births from the St Mary's Maternity Information System (SMMIS).

Main outcome measure Annual patterns of preterm birth and a comparison of risk by seasons.

Results Three studies from developing countries and three from developed countries reported a seasonal pattern of preterm birth. One study from the USA reported no seasonal pattern of preterm birth. No British studies were located. Rates of preterm birth in developed countries were highest twice a year (once in winter and again in summer). In London (SMMIS data set), however, preterm births peaked only once a year, in winter. Babies born in winter were 10% more likely to be preterm compared with those born in spring (OR 1.10, 95% CI 1.07–1.14).

Conclusion Establishing a seasonal pattern of birth can have important implications for the delivery of healthcare services. Most studies from both developed and developing countries support the existence of preterm birth seasonality. This study has shown that the seasonality of preterm births in this London-based cohort differs from other developed countries that have previously reported a seasonal pattern of preterm birth.

Keywords Preterm birth, seasonality, systematic review.

Please cite this paper as: Lee S, Steer P, Filippi V. Seasonal patterns and preterm birth: a systematic review of the literature and an analysis in a London-based cohort. BJOG 2006;113:1280–1288.

Introduction

Preterm birth, or birth before 37 completed weeks of gestation (less than 259 days), is one of the largest contributors to mortality in the neonatal period and infancy and to morbidity later in life.^{1,2} It accounts for 5–10% of all births in developed countries^{3,4}, and in the USA and Canada, rates of preterm birth are reported to be on the increase.^{5–7} Babies born between 32 and 36 weeks of gestation, which account for the majority of all preterm births, are approximately 3–15 times more likely to die within the first year of life compared with those born at term⁸; the economic costs for those who survive are very high.⁹ Long-term sequelae associated with

preterm birth include decreased motor and cognitive functioning and increased behavioural disorders, such as attention-deficit/hyperactivity disorder, when compared with children who are born at term.^{10,11}

Although much is known about the incidence and burden of preterm birth, its biological mechanisms are not well understood. An important step in reducing preterm birth and its consequences is likely to lie in furthering our understanding of the complex and often interrelated pathways that lead to preterm birth. One way in which this can be accomplished is by expanding our knowledge of how environmental factors affect or interact with biological mechanisms that lead to preterm birth.

The seasonality of preterm birth has been reported in both developed and developing countries, but it remains relatively unexplored.^{12–14} However, there may be seasons or months when the risk of preterm birth is greater or less than others, and this may shed light on the mechanisms underlying preterm birth. The number of preterm births may vary each month, but the risk depends on how many babies are *in utero* at any given time. Even if risk is not increased, there could be an increase in the absolute numbers of preterm births during any given month.

The causes of preterm birth seasonality may differ in developed versus developing countries. This study was undertaken to systematically establish countries in which a seasonal pattern of preterm birth has previously been reported and to present an analysis of the seasonal variability of preterm birth in a London-based cohort.

Methods

Systematic literature review

Before undertaking the review, efforts were made to locate any existing reviews on preterm birth seasonality by searching the Medline, the Embase and the Cochrane Library databases. No such existing reviews on preterm birth seasonality was found.

The review was restricted to published studies that defined preterm birth by length of gestation (rather than birthweight). Studies were identified through searches on Medline and Embase up to May 2005. Free text terms with wild cards and related medical subject headings (MeSH) were used. Keywords and phrases were categorised into two groups (Table 1). Terms within each group were combined using the 'or' operator to remove any duplicates before combining both groups using the 'and' operator to locate any relevant articles. All MeSH terms were exploded with all subheadings.

Studies from both developed and developing countries were included in this study. Studies returned by the search strategy were examined by three people to determine inclusion in the review (including S.J.L. and V.F.). Studies were excluded if they did not meet the following criteria:

- report proportions of preterm birth, or gestational age data from which proportions or risk of preterm birth could be calculated, by defined periods of time (e.g. by months or by 'seasons')
- span a study period of at least 1 year or compare data from different seasons within a year
- be written in English.

Supplementary searches using the MeSH terms, Infant-Very-Low-Birthweight and Infant-Low-Birthweight, were also conducted to ensure that no studies on preterm birth was inadvertently missed.

To maximise the chances of identifying all published studies on the seasonality of preterm birth, the references of

Table 1. Search strategy used in Medline and Embase

Group 1: Preterm birth	Group 2: Seasonality
Free text terms	Free text terms
Prematur*	Climat*
Gestation*	Season*
Pre?term	Weather*
MeSH terms	MeSH terms
Gestational age/infant, premature/labor, premature/	Climate/seasons/weather/ time factors/

included studies were examined to locate further articles that may have been missed, and the subject headings of all articles were scrutinised to determine whether the search strategy required modification.

Using standard appraisal questions,¹⁵ criteria were established *a priori* for assessing the quality of each study that met the inclusion criteria. To this end, specific information extracted from each study included information on research design, study population and definitions of preterm birth and seasonality that were provided.

Seasonality of preterm birth in a London-based cohort

Study population

Data from all pregnancies ($n = 509\,173$) from 1988 to 2000 in 18 maternity units in the Northwest Thames region of London, UK, were collected and entered into the St Mary's Maternity Information System (SMMIS) as part of routine clinical practice. Data were entered by a midwife or by a trained clerk and were subject to online validation and quality checks, with standard definitions for clinical measurements. The programmed checks reduced the likelihood of implausible values and ensured that the data in SMMIS were of high quality.¹⁶

This analysis included only singleton live births between 24 and 44 weeks of gestation weighing more than 200 g. Multiple births ($n = 13\,540$), stillbirths ($n = 1899$) and those with a birthweight less than 200 g ($n = 7$) were excluded from the study. As the lower boundary of 24 weeks represents the limit for live birth registration in the UK,¹⁷ births with missing gestational age data ($n = 911$) or with a gestational age less than 24 weeks ($n = 456$) were also excluded. Congenital anomalies ($n = 9340$) were excluded. An additional 255 records had a missing value for the number of infants born and were therefore not included, leaving 482 765 births for analysis.

Preterm birth was defined as birth at less than 37 weeks of gestation (i.e. less than 259 days). In the case of an irregular menstrual history or uncertain dates, the best antenatal assessment, taking dates, menstrual cycle and ultrasound(s) into account, was made.¹⁸

Climate in London

London has a temperate climate, which means that extremes of weather are rare.¹⁹ Four distinct seasons are recognised: spring (March through May), summer (June through August), Autumn (September through November) and winter (December through February). The average conditions in London over a year are shown in Figure 1.

Statistical analysis

Individual-level data (pregnancy records) were collapsed so that the final data set represented a time-series where each record represented a day during the study period ($n = 4749$ days over 13 years). Each record contained the number of births and number of preterm births per day. To determine which months experienced more preterm births, proportions were calculated using daily totals of live births, thereby adjusting for the different number of days each month. Because there were 4 leap years over the 13 year study period, 28.31 days was used for February (i.e. $[(29 \times 4) + (28 \times 9)]/13$).

To determine which seasons or months were associated with the highest risk of preterm birth, probabilities of preterm birth per 1000 at-risk fetuses were obtained. Denominators for probabilities were obtained by prospectively counting all continuing pregnancies between 24 and less than 37 weeks for each index day and then dividing the total number of preterm births each month (numerator) by the total number of fetuses at risk each month (denominator).

The proportions and probabilities were plotted for visual inspection. Any regularity in the series was drawn out using moving averages to remove some of the random variation and allow detection of any long-term patterns that may be of interest. This technique acted to 'smooth' the data by moving

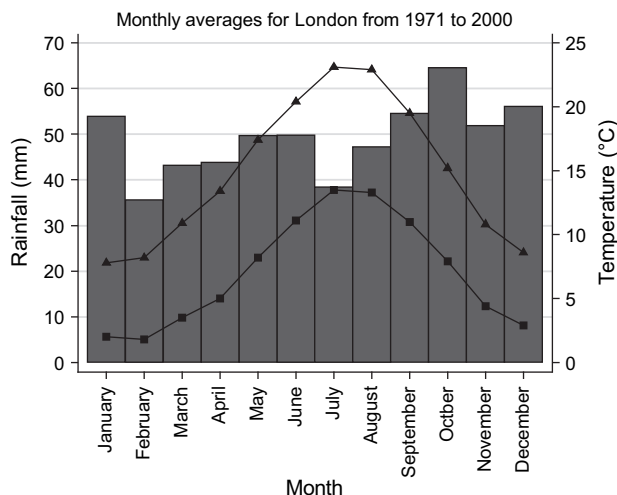


Figure 1. Weather averages for London from 1971 to 2000. Triangles, maximum mean monthly temperature; squares, minimum mean monthly temperature; bars, monthly mean rainfall amount (obtained from <http://www.met-office.gov.uk/education/data/climate/temperate/index.html>).

through each value and replacing it with the average of the value itself plus the values on either side of it. The number of values on either side to be included in the average was specified such that a moving average of five included the value itself plus two values on either side of the data point.

Associations by month and by season were evaluated using the chi-square test, and the magnitude of effect was calculated using odds ratios. A P value of less than 0.05 was considered significant.

Results

From the literature review

In total, the review located seven studies, published in eight papers. Two of the papers^{14,20} used the same data set for analysis; therefore, only the paper that was published first was included (Table 2). Three of the studies were from developing countries (one from Bangladesh, one from Zimbabwe and one from the Gambia) and four from developed countries (three from the USA and one from Japan). No British studies were located.

Using secondary data analysis, two developed countries reported similar preterm birth seasonality.^{14,21,22} In the USA and Japan, the pattern was annual, with the highest proportions and risk occurring twice a year: once in summer and again in winter. The lowest risk was observed in spring and autumn.

One study from the USA reported no seasonality of preterm birth.²³ This study compared preterm birth rates from May to October with those from November to April for 1983–85 in a northern Californian population and found no difference in spontaneous preterm birth proportions.

In the developing countries, all three studies that were located reported a seasonality of preterm birth using their own definitions of seasons. In Bangladesh,²⁴ the lowest proportion of preterm births occurred during winter (3.6%), followed by summer (5.5%) and monsoon (6.0%) season. The highest proportion of preterm births was observed in autumn (8.3%).

In Gambia, two seasons were reported.¹³ Preterm birth proportions were found to peak twice a year: once at the beginning of the hungry season in July and again towards the end of the hungry season in October.

Harare, Zimbabwe, was reported to experience four seasons.²⁵ Births in the early dry season were three times more likely to be preterm compared with those during the late rainy season (OR 2.9, 95% CI 1.65–5.2). No other seasonal differences in the risk of preterm birth between seasons were found.

The seasonality of preterm births in London

Proportions of preterm birth varied significantly by month ($\chi^2(11) = 96.57$, $P < 0.001$), with the highest proportion

Table 2. Summary of the studies located for the systematic review on the seasonality of preterm birth

Author, country	Design	Population	Methods	Preterm birth definition	Results
Matsuda and Kahyo ¹⁴ , Japan	Cross-sectional; secondary data analysis	All live singleton infants (n = 7 665 006) born between 1979 and 1983	Time-series analysis. Seasons: Spring (March to May), summer (June to August), autumn (September to November), winter (December to February). Results stratified by parity (first birth vs subsequent births) and by infant gender	Less than 37 weeks of gestation as reported on vital statistic records	First-born males: peaks in June and December and troughs in March and September/October. First-born females: peaks in July and December/January and troughs in March and October. Male-subsequent births: peaks in August and December and troughs in May and October. Female-subsequent births: peaks in August and December/January and troughs in May and October.
Keller and Nugent ²¹ , USA	Cross-sectional; secondary data analysis	Singleton births (n = 402 540) of white residents from 1967 to 1973	Number of pregnancies at risk was considered to be the total number of babies born each month for each gestational age band over the corresponding number of fetuses <i>in utero</i> each month	Live or stillbirth at 29–37 weeks of gestation, as reported on birth certificates	All pregnancy outcomes for each gestational group showed same seasonal pattern. Peak for probability of a preterm birth around August (approximately 59.8 preterm births/1000 at-risk fetuses) and December to January (approximately 58.7 preterm births/1000 at-risk fetuses). Low periods (troughs) in spring (April, approximately 55.5/1000 at-risk fetuses) and fall (October, 57.2/1000 at-risk fetuses). Total spread from peak to trough reported to be 8%

(continued)

Table 2. (Continued)

Author, country	Design	Population	Methods	Preterm birth definition	Results
Cooperstock and Wolfe ²² , USA	Cross-sectional; secondary data analysis	The $n = 928$ live, preterm and $n = 3651$ term births from 1959 to 1966	The number of preterm births each month was divided by the mean number of preterm births/month	Live born between 27 and 35 weeks of gestation as calculated from menstrual dates	Peak in September (144%) and trough in May (64%) for preterm births (adjusted for seasonal variation in fertility)
Konte et al. ²³ , USA	Cross-sectional	All women who delivered from May 1983 to June 1985, recruited between 21 and 24 weeks of gestation ($n = 9296$)	Preterm birth proportions for each period from May to October and for each period from November to April compared	Occurring between 140 and 259 days of gestation as a result of spontaneous preterm labour or spontaneous rupture of the membranes, except when delivery was indicated for maternal or fetal reasons unrelated to preterm labour or preterm labour treatment	No differences in spontaneous preterm birth proportions between May and October compared with that from April to November
Hort, ²⁴ Bangladesh	Cross-sectional	All live singleton babies born in 1983 or 1984 ($n = 1772$)	Seasons: Winter (December to February), summer (March to May), monsoon (June to August), autumn (September to November). Percentage of preterm births from among total births during each season	Less than 35 weeks of gestation, as assessed by Dubowitz method for babies below 2 kg	Winter, 3.6%; summer, 5.5%; monsoon, 6.0%; autumn, 8.3%
Rayco-Solon et al. ¹³ , Gambia	Cross-sectional; secondary data analysis	All live births with gestational age and birth weight data not missing ($n = 1916$) from 1976 to 2003	Annual rainy season from July to November. Logistic regression to assess dependence on seasonality of births	Less than 37 weeks of gestation as assessed by the Dubowitz scoring system	Proportion of preterm births showed an annual pattern with two peaks (July and October). Likelihood ratio, $\chi^2 = 20.07$, 6df, $P = 0.003$ for dependence on season of birth
Friis et al. ²⁵ , Zimbabwe	Cross-sectional; secondary analysis of randomised controlled trial data on micronutrient supplementation	Women between 22 and 36 weeks of gestation from 1996 to 1997 ($n = 1106$)	Seasons: early dry (June to August), late dry (September to November), early rainy (December to February) and late rainy (March to May). Logistic regression analysis (late rainy season used as reference group)	Less than 37 weeks of gestation as measured from first day of last menstruation or by fundus height	Early dry: OR 2.9, 95% CI 1.65–5.2; late dry: OR 0.94, 95% CI 0.51–1.72; early rainy: OR 1.38, 95% CI 0.75–2.52

(6.83%) occurring in January (Table 3). When plotted over the 13-year study period, preterm birth proportions were consistently highest during the winter and lowest during the summer (Figure 2). This was more clearly seen when the monthly proportions were averaged over 1 representative year (Figure 3).

The risk of preterm birth was found to be highest in November and December (Table 3). When examined by season, babies were 10% more likely to be born preterm in autumn or winter than in spring (OR 1.10, 95% CI 1.07–1.14 for both seasons). The risk of being born preterm in summer when compared with in spring did not reach statistical significance (OR 1.03, 95% CI 0.99–1.06).

The time of highest risk for preterm birth did not concur with the time when most births were occurring. Most births occurred towards the end of summer (Table 3). Despite the risk of preterm birth being highest in winters, the total number of preterm births each day was only slightly higher in winter than in summer.

Discussion

This study has shown that the seasonality of preterm births in this London-based cohort differs from other developed countries that have previously reported a seasonal pattern of preterm birth. While rates of preterm birth in countries such as Japan and the USA are at their highest twice a year (once in winter and again in summer), preterm births from the SMMIS data set peaked only once a year, in winter. Babies born in winter were 10% more likely to be preterm compared with those born in spring (OR 1.10, 95% CI 1.07–

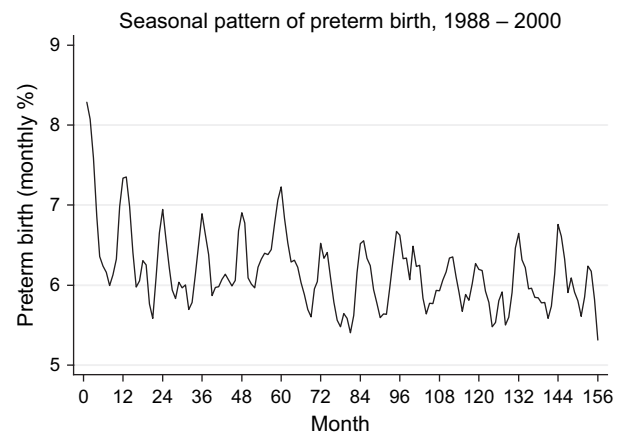


Figure 2. The seasonal pattern of monthly preterm birth proportions in SMMIS was consistent across the 13 study years. A moving average of 3 months was used for smoothing.

1.14). The summer increase in preterm birth seasonality seen in Japan and the USA was not apparent in the SMMIS population. Indeed, it was during this time of year that preterm births in the SMMIS data set were, on average, at their lowest. Throughout the year, the percentage of preterm births was as low as 5.6% and rose to as high as 6.8% in January.

Conflicting or slightly varying results between countries are most likely related to geographical, cultural and socio-economic differences between the populations studied. For example, in Japan, the winter peak (December and January) in preterm birth seasonality became more prominent in the northern part of the country and the summer peak (June and

Table 3. Preterm birth proportions and probabilities

Month	No. of births (%)	Births/day*	24–32 weeks**	32–36 weeks**	% preterm	P***
January	39 289 (8.14)	1267.39	11.84	74.68	6.83	0.73
February	36 705 (7.60)	1296.64	10.85	70.72	6.29	0.68
March	40 537 (8.40)	1307.65	10.90	70.10	6.19	0.66
April	39 330 (8.15)	1311.00	11.67	67.47	6.04	0.64
May	41 883 (8.68)	1351.06	11.48	69.23	5.97	0.65
June	41 327 (8.56)	1377.57	12.83	69.23	5.96	0.67
July	42 633 (8.83)	1375.26	11.68	70.71	5.99	0.68
August	41 711 (8.64)	1345.52	11.13	67.35	5.83	0.66
September	41 487 (8.59)	1382.90	11.83	65.63	5.60	0.68
October	40 725 (8.44)	1313.71	10.84	69.03	6.08	0.72
November	38 470 (7.97)	1282.33	12.10	71.70	6.53	0.76
December	38 668 (8.01)	1247.35	10.90	72.16	6.66	0.76
Total	482 765 (100)					

*Adjusted for the number of days each month; as there were 4 leap years during the 13-year study period, 28.31 days was used for February.

**Number of births/day.

***Probability per 1000 fetuses at risk.

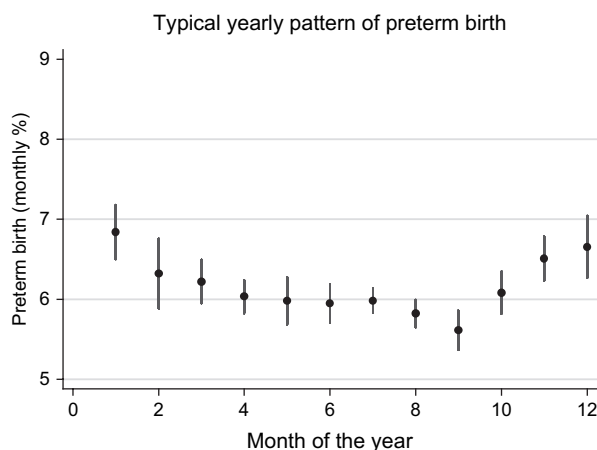


Figure 3. The seasonal pattern is more clearly seen when plotted over one 12-month period. In London-based hospitals, the largest percentage of preterm births occur in January (vertical bars represent 95% confidence intervals).

July) was more prominent in the southern part of the country.²⁰ This suggests a possible association between seasonal patterns and latitude.

One external environmental factor that is notably different in England compared with Japan and most parts of the USA is the range of temperature experienced by each country. The temperate climate in England means that the population is not exposed to the more extreme temperatures that are experienced throughout Japan and much of the USA. The mean summer temperature as measured by the weather monitoring station at Heathrow airport was 17.9°C for the study period (calculated with data from the British Atmospheric Data Centre [<http://badc.nerc.ac.uk/home/>]). In a study of geographical differences in the seasonal pattern of preterm birth in Japan, Matsuda and Kahyo²⁰ reported that mean summer temperatures for all 47 prefectures in Japan ranged from 19.5 to 27.4°C.

Hot tub use during pregnancy has been reported to be associated with an increased risk of miscarriage and neural tube defects,^{26,27} suggesting that one or more stages during fetal development are susceptible to harmful effects due to heat exposure. In one study of pregnant women, the rate of preterm labour from four different 1-week time periods was compared.²⁸ The 4 weeks selected represented the coldest and warmest weeks of winter and the coldest and warmest weeks of summer. Using the exact test for linear trend, preterm labour rates were reported to increase with an increasing heat-humidity index ($P < 0.002$), a measure combining relative humidity and temperature to give an indication of how hot it 'feels'. Therefore, while an association between high temperatures and an increased risk for preterm birth may explain the high rates of preterm birth observed in Japan and the USA, it may be that in Britain, pregnant women are

not exposed to the high temperatures needed to observe an effect.

Among developing countries, if a relationship between preterm birth and seasonality exists, it is likely to be due to factors different from those that operate in developed countries. For example, seasonal patterns in nutritional status and maternal weight loss have been implicated as factors responsible for the seasonal pattern of low birthweight in developing countries.^{29,30} These factors, however, are unlikely to play a large role in developed countries where seasonal food shortages are no longer prevalent and overall nutrition is generally good but where seasonality of low birthweight still exists.

A seasonality of conceptions, which may be driven by cultural practices and beliefs, could possibly create a seasonal pattern of preterm birth. In particular, if preterm birth rates were constant and conceptions seasonal, any preterm birth seasonality would be attributable to the seasonality of conceptions. Our study of the UK population, however, used both proportions and probabilities for analysis. While an analysis based on preterm birth proportions may be affected by a seasonality of conceptions, the fetuses at risk approach accounted for any seasonality of birth or conceptions by calculating its denominator prospectively. In fact, although the proportion of preterm births was higher in winter for this population, the total number of births per day was highest in summers. This resulted in the actual number of preterm births per day being only slightly higher in winter than in summer.

Keller and Nugent²¹ suggested that because different gestational age groups showed the same seasonal pattern of preterm birth, the seasonality was likely to be due to factors present at the time of birth rather than the time of conception. If one limits consideration to potential factors around the time of birth to explain a seasonal pattern of preterm birth, two possibilities arise. The first is that an associated risk factor, such as genital tract infections, shows seasonality^{31–33} and that this seasonality contributes to a seasonal pattern of preterm birth. Another possibility is that an association with some external environmental factor, such as sudden drops in barometric pressure or high levels of air pollution, act to 'trigger' an early initiation of labour, subsequently resulting in preterm birth. As many air pollutants and aspects of weather show distinct seasonal patterns, such a mechanism could also contribute to a preterm birth seasonality.

Publication bias is of concern in this systematic review. It is possible that the seasonality of preterm births has been investigated elsewhere, but no seasonality was found; therefore, the results were not subsequently published or published under a different heading so that these were not detected by the systematic review. The only study reporting no seasonality that was located was a study that mentioned this result in its discussion section only.²³

This review also only considered studies that were written in English. Previous studies have identified drawbacks and potential bias due to excluding studies written in other languages.^{34,35} There is also evidence, however, that language-restricted systematic reviews do not necessarily result in biased findings.³⁶

Conclusions

Investigating the seasonality of preterm birth can provide new insights useful in understanding, targeting and limiting the risk of preterm birth. Even a relatively small increase in the risk of morbidity at the higher preterm gestations may have a substantial impact on the attributable risk of morbidity (i.e. the risk of morbidity that can be attributed to being born preterm) because of the large numbers of babies affected. Therefore the small but measurable difference in preterm birth rate reported in this study, which resulted in a 10% increase in risk for preterm birth during winter, may have important implications. Morbidity and mortality due to preterm birth could potentially be reduced if hospitals and carers in the UK anticipated and prepared for the likelihood of babies being at higher risk of being born preterm during the winter months.

Acknowledgements

We would like to thank all the midwives, who entered the data, and the consultants at the participating maternity units, without whom this analysis would not have been possible. We would also like to thank Sara Thomas, who helped review the studies for the systematic review, and Noreen Maconochie and Shakoor Hajat, for their advice on the design of this study. ■

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